NEET UG 2027: Organic Reaction Mechanisms Handbook

Rigorous Step-by-Step Mechanisms, Intermediates, Kinetics & NEET Traps

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1 Electrophilic Aromatic Substitution (S_EAr / EAS)

Fundamental 3-Step EAS Pathway & The Arenium Ion (σ-Complex)

Electrophilic aromatic substitution is the hallmark reaction of electron-rich aromatic systems. The driving force is the preservation of aromatic resonance energy (≈ 150.6 kJ/mol in benzene). The reaction proceeds through three discrete elementary steps:

1. **Step 1: Generation of the Active Electrophile (E^+):** A Lewis acid or strong Brønsted acid polarizes or cleaves the reagent precursor to produce a potent cationic or polarized electrophilic species (E^+).
2. **Step 2: Formation of the Arenium Ion (σ-Complex / Wheland Intermediate) [RDS]:** The aromatic π-electron cloud attacks E^+. The attacked carbon rehybridizes from planar sp^2 to tetrahedral sp^3, temporarily disrupting aromaticity. The resulting carbocation is resonance-stabilized across the ortho and para positions:

$$C_6H_6 + E^+ \xrightarrow{\text{slow, RDS}} [C_6H_6E]^+ \quad \text{(Arenium Ion: positive charge delocalized over 5 ring carbons)} \tag{1}$$

3. **Step 3: Fast Deprotonation & Rearomatization:** The conjugate base abstracts the proton from the sp^3 carbon. The electron pair returns to the ring sextet, restoring full aromaticity (sp^2 rehybridization).

Kinetics, Energy Profile & Primary Kinetic Isotope Effect

1. Rate Law: Overall rate is governed by the second step (formation of the arenium ion):

$$\text{Rate} = k[\text{Ar}-\text{H}][E^+] \quad \text{(Second-order kinetics: first order in benzene, first order in electrophile)} \tag{2}$$

2. Kinetic Isotope Effect (k_H/k_D):

- **Nitration, Chlorination, Bromination, Alkylation:** $k_H/k_D \approx 1.0$. The C − H bond cleavage occurs in Step 3 (fast step) after the rate-determining transition state. Hence, C_6H_6, C_6D_6, and C_6T_6 react at identical rates.
- **Sulfonation & Iodination:** $k_H/k_D > 1.0$ ($k_H/k_D \approx 2-3$ for sulfonation; ≈ 4 for iodination). Here, Step 3 or intermediate decomposition has a barrier comparable to Step 2; Step 3 is reversible and partially rate-limiting.

1.1 Nitration of Aromatic Rings

Nitration Mechanism & Nitronium Ion Generation

Nitration is carried out using a "nitrating mixture" consisting of concentrated nitric acid (HNO_3) and concentrated sulfuric acid (H_2SO_4) at $323-333$ K:

1. **Generation of Nitronium Ion (NO_2^+):** Sulfuric acid ($pK_a \approx -3$) is a stronger acid than nitric acid ($pK_a \approx -1.4$). Nitric acid acts as a Brønsted-Lowry **base**, accepting a proton on its hydroxyl oxygen:

$$HNO_3 + H_2SO_4 \rightleftharpoons H_2\overset{+}{O} - NO_2 + HSO_4^- \quad \text{(Protonation of nitric acid)} \tag{3}$$

$$H_2\overset{+}{O} - NO_2 \rightleftharpoons NO_2^+ + H_2O \quad \text{(Heterolytic loss of water)} \tag{4}$$

$$H_2O + H_2SO_4 \rightleftharpoons H_3O^+ + HSO_4^- \quad \text{(Protonation of water by second } H_2SO_4\text{)} \tag{5}$$

$$\text{Net: } HNO_3 + 2H_2SO_4 \rightleftharpoons NO_2^+ + H_3O^+ + 2HSO_4^- \quad (NO_2^+ \text{ is linear, } sp\text{-hybridized}) \tag{6}$$

2. **Electrophilic Attack (RDS):** $C_6H_6 + NO_2^+ \xrightarrow{\text{slow}} [C_6H_6NO_2]^+$ (σ-complex).
3. **Rearomatization:** $[C_6H_6NO_2]^+ + HSO_4^- \xrightarrow{\text{fast}} C_6H_5NO_2 + H_2SO_4$.

NCERT Critical Extraction: Nitration of Aniline & Protecting Groups

Direct nitration of aniline with conc. HNO_3/H_2SO_4 gives an anomalous product distribution:

$$\text{Aniline} \xrightarrow[288\ \text{K}]{\text{conc. } HNO_3+H_2SO_4} p\text{-nitroaniline } (51\%) + \textbf{m-nitroaniline } (\textbf{47}\%) + o\text{-nitroaniline } (2\%) \tag{7}$$

Mechanistic Cause: In strongly acidic medium, the highly activating $-NH_2$ group ($+M$) is protonated to form the anilinium ion ($-NH_3^+$), which is a powerful $-I$ deactivating group that directs meta. **Synthetic Solution (Acetylation Protection):** Aniline is first protected with acetic anhydride (Ac_2O) in pyridine to form acetanilide ($Ph-NHCOCH_3$). The lone pair on nitrogen is delocalized into the carbonyl group by resonance, reducing its basicity (preventing protonation) and modulating activating strength. Nitration then gives exclusively **para-nitroacetanilide**, followed by acid/base hydrolysis to yield pure **para-nitroaniline**.

1.2 Halogenation of Benzene

Halogenation Pathways & Lewis Acid Catalysis

- **Chlorination / Bromination:** Benzene reacts with Cl_2 or Br_2 in the presence of anhydrous Lewis acids ($FeCl_3, AlCl_3, FeBr_3$):

$$Cl-Cl + FeCl_3 \rightleftharpoons [Cl^{\delta+}\cdots Cl\cdots {FeCl_3}^{\delta-}] \rightarrow Cl^+ + [FeCl_4]^- \tag{8}$$

  Electrophilic Cl^+ attacks the ring to form the chlorocyclohexadienyl cation, followed by $[FeCl_4]^-$ deprotonation yielding chlorobenzene and regenerating $FeCl_3 + HCl$.
- **Fluorination:** Direct fluorination is violently exothermic ($\Delta H \ll 0$) leading to explosive rupture of $C-C$ bonds. Fluorobenzene is prepared via the Balz-Schiemann reaction: $Ar-NH_2 \xrightarrow{NaNO_2/HCl} Ar-N_2^+Cl^- \xrightarrow{HBF_4} Ar-N_2^+BF_4^- \xrightarrow{\Delta} Ar-F + BF_3 + N_2\uparrow$.
- **Iodination:** Reversible due to formation of reducing agent HI:

$$C_6H_6 + I_2 \rightleftharpoons C_6H_5I + HI \tag{9}$$

  To drive equilibrium forward, an in situ oxidizing agent (HNO_3, HIO_3, or HgO) is added to oxidize HI to I_2: $5HI + HIO_3 \rightarrow 3I_2 + 3H_2O$.

1.3 Friedel-Crafts Alkylation & Acylation

Friedel-Crafts Alkylation: Mechanism & Classical Carbocation Rearrangements

1. Mechanism:

$$\mathrm{R-Cl + AlCl_3 \rightleftharpoons R^+[AlCl_4]^-} \quad \text{(Carbocation intermediate generated)} \tag{10}$$

$$\mathrm{C_6H_6 + R^+ \xrightarrow{\text{slow, RDS}} [C_6H_6R]^+ \xrightarrow{[AlCl_4]^-} C_6H_5R + HCl + AlCl_3} \tag{11}$$

2. Carbocation Rearrangement Trap: Whenever 1° alkyl halides with ≥ 3 carbons are used, the primary carbocation rearranges via 1,2-hydride or 1,2-methide shift to a more stable 2° or 3° carbocation prior to electrophilic attack:

$$\mathrm{C_6H_6 + CH_3CH_2CH_2Cl \xrightarrow{\text{anhyd. } AlCl_3}} \textbf{ Cumene (isopropylbenzene, 65–70\%)} + n\text{-propylbenzene (30–35\%)} \tag{12}$$

Mechanism of shift: $\mathrm{CH_3-CH_2-\overset{+}{C}H_2 \xrightarrow{\text{1,2-hydride shift}} CH_3-\overset{+}{C}H-CH_3}$ (2° carbocation is stabilized by 6 α-H vs 2 α-H).

Four Cardinal NEET Limitations of Friedel-Crafts Alkylation

1. **Polyalkylation Problem:** The introduced alkyl group ($-$R) is electron-donating ($+I, +H$), making the monoalkylated product more reactive toward further EAS than benzene itself. Hence, ortho/para dialkyl and polyalkyl benzenes form readily.
2. **Carbocation Rearrangement:** Straight unbranched alkyl benzenes cannot be synthesized cleanly in high yield by direct alkylation.
3. **Deactivated Ring Immunity:** Rings carrying moderate to strong electron-withdrawing groups ($-M/-I$ like $-NO_2, -CF_3, -SO_3H, -COOH, -COOR, -CN, -CHO$) do **not** undergo Friedel-Crafts reactions. Nitrobenzene is routinely used as an inert solvent for Friedel-Crafts reactions!
4. **Aniline / Phenol Lewis Acid Failure:** Aniline and phenol do not undergo Friedel-Crafts alkylation/acylation cleanly because the Lewis acid ($AlCl_3$) forms a strong coordinate complex with the lone pair on $-NH_2$ or $-OH$:

$$\mathrm{C_6H_5NH_2 + AlCl_3 \rightarrow C_6H_5-\overset{+}{N}H_2-\overset{-}{A}lCl_3} \tag{13}$$

This converts the powerfully activating $-NH_2$ into a strongly deactivating $-I$ group ($-\overset{+}{N}H_2-AlCl_3^-$), completely terminating EAS.

Friedel-Crafts Acylation: Acylium Ion Intermediate & Remedy for Polyalkylation

Acylation avoids both the rearrangement and polyalkylation defects of alkylation:

1. **Generation of the Acylium Ion:**

$$\mathrm{R-CO-Cl + AlCl_3 \rightleftharpoons [R-\overset{+}{C}=O \longleftrightarrow R-C\equiv\overset{+}{O}] + [AlCl_4]^-} \tag{14}$$

The acylium ion is exceptionally stable because all atoms possess a complete octet in the $R-C\equiv O^+$ canonical form. Consequently, it **never undergoes carbocation rearrangement**!

2. **Attack & Single Substitution:** $\mathrm{C_6H_6 + RCO^+ \xrightarrow{RDS} [C_6H_6COR]^+ \rightarrow C_6H_5COR + HCl}$.
3. **No Polyacylation:** The introduced acyl group ($-$COR) is strongly electron-withdrawing ($-M, -I$), deactivating the product ring against further electrophilic attack.
4. **Synthesis of Unbranched Alkylbenzenes:** The acyl product is subsequently reduced to an unbranched alkyl chain via either:
   - **Clemmensen Reduction:** $\mathrm{Ar-CO-R \xrightarrow{Zn-Hg/conc.\ HCl} Ar-CH_2-R}$ (for acid-stable substrates).
   - **Wolff-Kishner Reduction:** $\mathrm{Ar-CO-R \xrightarrow[KOH,ethylene\ glycol,\ \Delta]{NH_2NH_2} Ar-CH_2-R + N_2\uparrow}$ (for base-stable substrates).

1.4 Orientation & Reactivity: Arenium Ion Resonance Analysis

Arenium Canonical Analysis: Ortho/Para vs Meta Directing Effects

The directing effect is governed entirely by the relative thermodynamic stability of the arenium intermediate for ortho, meta, and para electrophilic attack:

- **Ortho/Para Directing Groups ($+M > -I$, e.g., $-\text{OH}, -\text{OR}, -\text{NH}_2, -\text{NHCOR}$ or $+I/+H$, e.g., $-\text{CH}_3$):** Electrophilic attack at the ortho or para position places the positive charge directly on the ring carbon attached to the substituent. If the substituent has a lone pair ($+M$), it donates electrons into the ring, forming an especially stable **fourth resonance structure** where every atom has a complete octet.
- **Meta Directing Groups ($-M/-I$, e.g., $-\text{NO}_2, -\text{CN}, -\text{CHO}, -\text{COOH}, -\text{SO}_3\text{H}$):** Ortho and para attack generates a canonical form where the positive charge is placed directly adjacent to the positively polarized substituent carbon ($\text{C}^+ - \overset{\delta+}{\text{C}} = \text{O}$ or $\text{C}^+ - \overset{+}{\text{N}}\text{O}_2^-$), resulting in catastrophic electrostatic repulsion. Meta attack never places the positive charge directly on the ipso carbon, thereby minimizing destabilization. Hence, meta attack has a lower activation barrier than ortho/para attack.

The Halogen Paradox in EAS: Deactivating yet Ortho/Para Directing

Halogens ($-\text{F}, -\text{Cl}, -\text{Br}, -\text{I}$) are unique in organic chemistry:

- **Reactivity (Deactivating):** The inductive effect ($-I$) of the highly electronegative halogen strongly withdraws σ-electron density from the entire π-system, reducing overall ring electron density. Hence, halobenzenes react **slower** than benzene in all EAS reactions.
- **Orientation (Ortho/Para):** When attack occurs at the ortho or para position, the halogen lone pair can be donated via resonance ($+M$) to form a fourth canonical contributor ($= \overset{+}{\text{X}}$) with complete valence octets. This stabilization is absent for meta attack.
- **Conclusion:** Inductive effect controls reactivity (deactivating), while resonance effect controls orientation (o/p-directing).

2 Nucleophilic Aliphatic Substitution (S_N1 vs S_N2)

2.1 The S_N1 Reaction Pathway

Two-Step Unimolecular Substitution (S_N1)

The S_N1 mechanism operates through a stepwise dissociation-addition mechanism:

1. **Step 1: Heterolysis to form Carbocation Intermediate [RDS]:**

$$\mathrm{R-X} \xrightarrow[\text{slow, RDS}]{k_1} \mathrm{R^+ + X^-} \quad \text{(Planar, trigonal } sp^2 \text{ hybridized carbocation intermediate)} \tag{15}$$

   The rate depends solely on the concentration of the alkyl halide.

2. **Step 2: Nucleophilic Attack [Fast]:**

$$\mathrm{R^+ + Nu^-} \xrightarrow[\text{fast}]{k_2} \mathrm{R-Nu} \tag{16}$$

   The nucleophile can attack either the front face or back face of the unhindered planar carbocation.

S_N1 Governing Equations, Stereochemistry & Solvation

Rate Law: $\text{Rate} = k[\mathrm{R-X}] \quad (\text{Order} = 1, \text{Molecularity} = 1)$

Substrate Hierarchy: $\text{Benzylic} \approx \text{Allylic} \approx 3^\circ > 2^\circ \gg 1^\circ > \mathrm{CH_3X}$

Solvent Acceleration: Favored by polar protic solvents ($\mathrm{H_2O}, \mathrm{MeOH}, \mathrm{EtOH}, \mathrm{AcOH}$) with high dielectric constant (ϵ), which solvate and stabilize both the carbocation and leaving group anion via hydrogen bonding.

Stereochemical Outcome (Racemization with Partial Inversion): The leaving group $\mathrm{X^-}$ does not diffuse away instantaneously; it forms an **intimate ion pair** ($\mathrm{R^+ \cdots X^-}$) that partially shields the front face. Consequently, backside attack occurs more frequently than frontside attack, leading to **net partial racemization with predominant inversion** (typically $\approx 50-70\%$ inversion $+30-50\%$ retention, corresponding to $60-100\%$ racemic mixture with $0-40\%$ net optical inversion).

Carbocation Rearrangements in S_N1

Because a free carbocation is formed in Step 1, S_N1 reactions are prone to Wagner-Meerwein rearrangements:

- **1,2-Hydride Shift:** 2-chloro-3-methylbutane solvolyzed in ethanol yields predominantly 2-ethoxy-2-methylbutane (3° product via hydride shift from 2° carbocation).
- **1,2-Methyl Shift:** Neopentyl bromide solvolyzed in water yields 2-methylbutan-2-ol (3° carbocation via methide shift).
- **Ring Expansion:** Cyclobutylmethyl halide yields cyclopentanol derivatives to relieve 90° ring angle strain.

2.2 The S_N2 Reaction Pathway

One-Step Bimolecular Concerted Substitution (S_N2)

The S_N2 mechanism occurs in a single, concerted step without any reactive intermediate:

1. The nucleophile approaches the electrophilic carbon from the **backside** (180° relative to the C − X leaving bond).
2. The reaction passes through a single, pentacoordinate, trigonal bipyramidal transition state:

$$\mathrm{Nu^- + R - CH_2 - X \xrightarrow{slow} \left[Nu^{\delta-} \cdots CH_2R \cdots X^{\delta-}\right]^{\ddagger} \xrightarrow{fast} Nu - CH_2 - R + X^-} \tag{17}$$

The central carbon is approximately sp^2 hybridized with three equatorial bonds, while the nucleophile and leaving group share a collinear p-orbital axis at 180°.

S_N2 Rate Law, Steric Effects & Walden Inversion

Rate Law: $\mathrm{Rate} = k[\mathrm{R-X}][\mathrm{Nu^-}]$ (Order = 2, Molecularity = 2)

Substrate Reactivity: $\mathrm{CH_3X} > 1^\circ > 2^\circ \ggg 3^\circ$ (Steric hindrance is dominant!)

Relative Rates: $\mathrm{CH_3Br}$ (1200) > EtBr (40) > i-PrBr (1) > t-BuBr (0.001)

Stereochemical Outcome (100% Walden Inversion): Backside attack forces the three equatorial substituents to flip through a planar transition state, resembling an umbrella blown inside out by a gust of wind. This results in complete inversion of configuration at the stereocenter ($R \to S$ or $S \to R$, depending on Cahn-Ingold-Prelog priorities).

The Neopentyl Halide S_N2 Trap & Solvent Effects

- **Neopentyl Halide Inertness:** Neopentyl bromide, $(\mathrm{CH_3})_3\mathrm{C-CH_2-Br}$, is a primary alkyl halide (1°). However, its S_N2 rate is 10^{-5} times slower than ethyl bromide! The three bulky methyl groups on the β-carbon completely block backside access to the electrophilic α-carbon.
- **Polar Aprotic Solvents Accelerate S_N2:** Solvents such as DMSO (dimethyl sulfoxide), DMF (N,N-dimethylformamide), acetone, and acetonitrile coordinate strongly to cations via their electronegative oxygen atoms, but cannot hydrogen-bond to nucleophilic anions. The anions remain "naked" and desolvated, boosting their nucleophilicity by up to 10^5-fold.

2.3 Master Diagnostic: S_N1 vs S_N2 vs S_Ni

Parameter	S_N1 Mechanism	S_N2 Mechanism
Number of Steps	Two steps (carbocation intermediate).	One concerted step (pentacoordinate transition state).
Kinetic Rate Law	$\mathrm{Rate} = k[\mathrm{R-X}]$ (1st order, unimolecular).	$\mathrm{Rate} = k[\mathrm{R-X}][\mathrm{Nu^-}]$ (2nd order, bimolecular).
Substrate Preference	$3^\circ > 2^\circ \gg 1^\circ$; benzylic ≈ allylic > 3°.	$\mathrm{CH_3X} > 1^\circ > 2^\circ \ggg 3^\circ$ (neopentyl extremely slow).
Nucleophile	Weak nucleophile is sufficient ($\mathrm{H_2O}$, ROH).	Strong, unhindered nucleophile required ($\mathrm{OH^-}, \mathrm{RO^-}, \mathrm{CN^-}, \mathrm{I^-}$).
Solvent Requirement	Polar protic solvent ($\mathrm{H_2O}$, MeOH, EtOH) stabilizes ions.	Polar aprotic solvent (DMSO, DMF, acetone) desolvates $\mathrm{Nu^-}$.
Stereochemistry	Racemization with slight excess of inversion.	100% Walden Inversion of configuration.
Rearrangements	Readily observed (hydride/alkyl shifts, ring expansion).	Completely absent (no carbocation intermediate).
Leaving Group Order	$\mathrm{I^- > Br^- > Cl^- \gg F^-}$ (governed by C − X bond energy).	$\mathrm{I^- > Br^- > Cl^- \gg F^-}$ (governed by leaving group stability).

S_Ni Mechanism (Substitution Nucleophilic Internal): Thionyl Chloride Reaction

When an alcohol reacts with thionyl chloride ($\mathrm{SOCl_2}$):

- **In Ether (No Pyridine):** $\mathrm{R-OH + SOCl_2 \xrightarrow{ether} R-Cl + SO_2 \uparrow + HCl \uparrow}$. Proceeds via an internal chlorosulfite intermediate (R − O − SOCl) with intramolecular delivery of chlorine from the same face → **Complete Retention of Configuration (S_Ni)**.
- **In Pyridine Medium:** Pyridine neutralizes HCl to form pyridinium chloride ($\mathrm{C_5H_5NH^+Cl^-}$), releasing free chloride ions ($\mathrm{Cl^-}$). These chloride ions attack the chlorosulfite from the backside → **Complete Inversion of Configuration (S_N2)**.

3 Elimination Reactions ($E1$ vs $E2$) & Dehydration of Alcohols

3.1 $E1$ Elimination Mechanism

Unimolecular Elimination ($E1$): Steps & Thermodynamic Drive

$E1$ elimination proceeds through ionization to a planar carbocation, identical to Step 1 of S_N1:

1. **Step 1 (RDS):** $\mathrm{R-X} \xrightarrow{\text{slow, RDS}} \mathrm{R^+ + X^-}$.
2. **Step 2 (Fast):** A weak base ($\mathrm{H_2O}, \mathrm{EtOH}$) abstracts a β-hydrogen from the carbocation:

$$\mathrm{R-CH_2-\overset{+}{C}H-CH_3 + B: \xrightarrow{fast} R-CH=CH-CH_3 + BH^+} \tag{18}$$

3. **Thermodynamic Competition (S_N1 vs $E1$):** Because Step 1 is shared, S_N1 and $E1$ compete directly. Elimination yields 2 product molecules from 1 reactant ($\Delta S^\circ > 0$). In the Gibbs equation $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$, higher temperatures increase the magnitude of $-T\Delta S^\circ$, favoring $E1$ **elimination over S_N1 substitution at elevated temperatures**.
4. **Regiochemistry:** Governed by **Saytzeff's Rule** (the more substituted, thermodynamically stable alkene with more hyperconjugative α-hydrogens is the major product; *trans* > *cis*).

3.2 $E2$ Elimination: Anti-Periplanar Geometry & Regioselectivity

Bimolecular Elimination ($E2$): Stereoelectronic Rigidity

$E2$ elimination occurs in a concerted manner upon treatment with a strong base (such as alcoholic KOH or $\mathrm{NaOCH_2CH_3}$):

$$\text{Rate} = k[\mathrm{R-X}][\text{Base}] \quad (\text{Order} = 2, \text{Molecularity} = 2) \tag{19}$$

Substrate Order: $3^\circ > 2^\circ > 1^\circ$ (developing double bond in transition state is stabilized by alkyl groups).

Stereoelectronic Requirement (Anti-Periplanar Geometry): The $\mathrm{C_\beta - H}$ bond and $\mathrm{C_\alpha - X}$ bond must lie in the same plane at a dihedral angle of 180° (**anti-coplanar**). This orientation allows the electron pair from the breaking $\mathrm{C-H}$ bond to smoothly overlap into the developing π-bond while minimizing electron repulsion between the incoming base and departing halide.

Saytzeff (Zaitsev) vs Hofmann Regioselectivity Matrix

1. Saytzeff Rule (Thermodynamic Major Product): With normal unhindered strong bases ($\mathrm{OH^-}, \mathrm{OEt^-}$) and good leaving groups ($\mathrm{I^-}, \mathrm{Br^-}, \mathrm{Cl^-}$), the transition state resembles the product alkene. The more substituted, hyperconjugated alkene is formed as the major product:

$$\mathrm{CH_3-CH_2-CH(Br)-CH_3} \xrightarrow{\text{alc. KOH},\Delta} \textbf{But-2-ene (81\%, Saytzeff)} + \text{But-1-ene (19\%, Hofmann)} \tag{20}$$

2. Hofmann Rule (Kinetic Major Product): The less substituted, terminal alkene is obtained as the major product under two distinct scenarios:

- **Scenario A: Poor Leaving Group (e.g., Alkyl Fluorides, Quaternary Ammonium Hydroxides):** Because $\mathrm{F^-}$ or $-\mathrm{NR_3^+}$ is an extremely poor leaving group, the base begins abstracting the β-proton before the $\mathrm{C-X}$ bond breaks significantly. This develops substantial **carbanionic character** in the transition state. Since primary carbanions are more stable than secondary carbanions ($1^\circ > 2^\circ > 3^\circ$), the base preferentially deprotonates the more acidic, least hindered methyl carbon:

$$\mathrm{CH_3CH_2CH(F)CH_3} \xrightarrow{\mathrm{EtO^-Na^+}} \textbf{But-1-ene (80\%, Hofmann)} + \text{But-2-ene (20\%)} \tag{21}$$

- **Scenario B: Sterically Bulky Bases (e.g., Potassium tert-butoxide, *t*-BuOK or LDA):** The bulky tert-butoxide anion cannot access the sterically congested internal secondary β-hydrogen, forcing abstraction of the easily accessible, unhindered terminal β-hydrogen:

$$\mathrm{CH_3CH_2CH(Br)CH_3} \xrightarrow{(CH_3)_3CO^-K^+} \textbf{But-1-ene (73\%, Hofmann)} + \text{But-2-ene (27\%)} \tag{22}$$

3.3 Acid-Catalyzed Dehydration of Alcohols

Dehydration of Alcohols: Step-by-Step Mechanism

Dehydration of alcohols to alkenes is an acid-catalyzed elimination ($E1$ for 2° and 3° alcohols; $E2$ for 1° alcohols):

1. **Step 1: Protonation of Hydroxyl Group (Fast Equilibrium):**

$$\mathrm{R-OH + H^+ \rightleftharpoons R-\overset{+}{O}H_2} \quad \text{(Converts poor leaving group } -\mathrm{OH} \text{ into excellent leaving group } \mathrm{H_2O}) \tag{23}$$

2. **Step 2: Heterolysis to form Carbocation [RDS]:**

$$\mathrm{R-\overset{+}{O}H_2 \xrightarrow{\text{slow, RDS}} R^+ + H_2O} \tag{24}$$

3. **Step 3: Carbocation Rearrangement (if applicable):** 1,2-hydride or 1,2-methyl shifts to yield the most stable carbocation.
4. **Step 4: Deprotonation (Fast):** Water or $\mathrm{HSO_4^-}$ abstracts β-H to form the Saytzeff alkene.

NCERT Class 12 Benchmark: Reaction Conditions & Temperature Trap

- **Reactivity Hierarchy:** 3° alcohol > 2° alcohol > 1° alcohol (parallels carbocation stability: 3° > 2° > 1°).
- **Ethanol Conditions:**
  - At 443 K(170°C) with conc. $\mathrm{H_2SO_4}$: **Ethene** ($\mathrm{CH_2 = CH_2}$) forms via intramolecular dehydration.
  - At 413 K(140°C) with excess ethanol: **Diethyl ether** ($\mathrm{C_2H_5OC_2H_5}$) forms via bimolecular nucleophilic substitution (S_N2).
- 2° **Alcohol (Propan-2-ol):** Requires only 85% $\mathrm{H_3PO_4}$ at 440 K.
- 3° **Alcohol (2-Methylpropan-2-ol):** Requires only 20% $\mathrm{H_2SO_4}$ at mild temperature of 358 K.

4 Carbonyl Chemistry Mechanisms

4.1 Nucleophilic Addition to Aldehydes & Ketones (Ad_N)

Carbonyl Electrophilicity & Trajectory of Nucleophilic Attack

The carbonyl group ($>C=O$) is strongly polarized due to the high electronegativity of oxygen ($\chi_O = 3.5$ vs $\chi_C = 2.5$). The carbonyl carbon is planar sp^2 with an empty π^* LUMO.

1. **Burgi-Dunitz Angle:** The nucleophile attacks the electrophilic carbon at an angle of $\approx 107°$ relative to the $C=O$ bond axis, maximizing overlap with π^* while minimizing repulsion with oxygen lone pairs.
2. **Reactivity Sequence:**

$$HCHO > CH_3CHO > CH_3COCH_3 > C_6H_5CHO > C_6H_5COCH_3 > C_6H_5COC_6H_5 \tag{25}$$

Governing Factors:

- **Electronic Effect:** Electron-donating alkyl groups ($+I$) disperse the partial positive charge on carbon ($\delta+$), reducing its electrophilicity. Conjugation with an aromatic ring (resonance) stabilizes the ground state carbonyl, diminishing reactivity.
- **Steric Effect:** As substituents on the carbonyl carbon increase in bulk, transition state crowding during conversion from trigonal planar (120°) to tetrahedral (109.5°) increases, raising the activation energy.

Key Nucleophilic Addition Mechanisms

1. Addition of HCN (Cyanohydrin Formation):

- Pure HCN is a weak acid ($pK_a = 9.2$) and reacts extremely slowly. The reaction is base-catalyzed:

$$HCN + OH^- \rightleftharpoons CN^- + H_2O \quad \text{(Generates strong nucleophile } CN^-\text{)} \tag{26}$$

- Step 1 (RDS): Attack of CN^- on carbonyl carbon: $R_2C=O + CN^- \xrightarrow{\text{slow}} R_2C(CN)O^-$.
- Step 2 (Fast): Proton transfer: $R_2C(CN)O^- + HCN \xrightarrow{\text{fast}} R_2C(CN)OH + CN^-$.

2. Addition of Sodium Bisulfite ($NaHSO_3$): Sulfur is more nucleophilic than oxygen. The bisulfite nucleophile attacks via sulfur to form a tetrahedral intermediate, followed by proton transfer to give a crystalline, water-soluble bisulfite adduct:

$$RCHO + NaHSO_3 \rightleftharpoons R-CH(OH)SO_3^-Na^+ \quad \text{(White crystalline precipitate)} \tag{27}$$

Used for the separation and purification of aldehydes and aliphatic methyl ketones (sterically hindered ketones do not react). **3. Addition of Ammonia Derivatives (H_2N-Z):** Reagents: $Z=-OH$ (hydroxylamine $\rightarrow$ oxime), $-NH_2$ (hydrazine $\rightarrow$ hydrazone), $-NHC_6H_5$ (phenylhydrazine $\rightarrow$ phenylhydrazone), $-NH-C_6H_3(NO_2)_2$ (2,4-DNP $\rightarrow$ orange Brady's precipitate), $-NHCONH_2$ (semicarbazide $\rightarrow$ semicarbazone).

- **Optimum pH Control (pH ≈ 4.5):** If medium is too acidic (pH < 3), the amine reagent is completely protonated to $H_3\overset{+}{N}-Z$, destroying its nucleophilicity. If medium is too basic (pH > 6), acid-catalyzed dehydration of the carbinolamine intermediate is too slow.

4.2 Aldol Condensation & Cross-Aldol Mechanism

Aldol Condensation: Step-by-Step Mechanism

Aldol condensation occurs in aldehydes and ketones possessing at least one α-hydrogen when treated with dilute alkali ($NaOH, Ba(OH)_2, Na_2CO_3$):

1. **Step 1: Enolate Generation (Acid-Base Equilibrium):** Hydroxide abstracts an acidic α-proton ($pK_a \approx 19-20$, acidified by resonance stabilization and $-I$ of carbonyl):

$$CH_3CHO + OH^- \rightleftharpoons [^-CH_2CHO \longleftrightarrow CH_2=CHO^-] + H_2O \quad \text{(Enolate carbanion)} \tag{28}$$

2. **Step 2: Nucleophilic Addition [RDS]:** The enolate carbanion attacks the electrophilic carbonyl carbon of a second un-ionized aldehyde molecule:

$$CH_3CHO + CH_2 = CHO^- \xrightarrow{\text{slow, RDS}} CH_3 - CH(O^-) - CH_2 - CHO \quad \text{(Alkoxide intermediate)} \tag{29}$$

3. **Step 3: Protonation to Form β-Hydroxyaldehyde (Aldol):**

$$CH_3 - CH(O^-) - CH_2 - CHO + H_2O \rightleftharpoons CH_3 - CH(OH) - CH_2 - CHO \text{ (3-hydroxybutanal)} + OH^- \tag{30}$$

4. **Step 4: Condensation / Dehydration ($E1cB$ Mechanism):** Upon heating (Δ), the aldol eliminates water. Hydroxide abstracts the remaining acidic α-hydrogen to form an enolate, followed by expulsion of hydroxide ion (OH^-, poor leaving group expelled due to conjugation driving force):

$$CH_3 - CH(OH) - CH_2 - CHO \xrightarrow[\Delta]{OH^-} CH_3 - CH = CH - CHO \text{ (Crotonaldehyde)} + H_2O \tag{31}$$

The resulting α,β-unsaturated enal/enone is thermodynamically stabilized by extended conjugation between the C = C double bond and the carbonyl group.

Cross-Aldol Condensation: Product Complexity & Directed Syntheses

- **Mixture of Two Aldehydes with α-H:** Reacting ethanal (CH_3CHO) with propanal (CH_3CH_2CHO) gives a complex mixture of **four** condensation products:
  1. Self-condensation of ethanal: But-2-enal.
  2. Self-condensation of propanal: 2-Methylpent-2-enal.
  3. Cross-condensation (ethanal enolate + propanal electrophile): Pent-2-enal.
  4. Cross-condensation (propanal enolate + ethanal electrophile): 2-Methylbut-2-enal.
- **Claisen-Schmidt Condensation (Directed Cross-Aldol):** To obtain a single clean product, one carbonyl component must lack α-hydrogens (e.g., benzaldehyde, C_6H_5CHO, or formaldehyde, HCHO). The partner with α-hydrogens acts exclusively as the nucleophile:

$$PhCHO + CH_3COCH_3 \xrightarrow[293\text{–}303\ K]{\text{dil. NaOH}} Ph - CH = CH - COCH_3 \text{ (Benzalacetone)} + H_2O \tag{32}$$

4.3 Cannizzaro Reaction & Hydride Transfer Mechanism

Cannizzaro Disproportionation: Step-by-Step Hydride Shift

Aldehydes that lack α-hydrogens (e.g., $HCHO, C_6H_5CHO, (CH_3)_3CCHO$) cannot form an enolate. Upon heating with concentrated alkali (50% NaOH or KOH), they undergo self-oxidation-reduction (disproportionation):

1. **Step 1: Rapid Nucleophilic Addition of Hydroxide:**

$$Ar - CHO + OH^- \rightleftharpoons Ar - CH(OH)O^- \quad \text{(Monoanion hydrate)} \tag{33}$$

2. **Step 2: Second Deprotonation in Strongly Basic Media:**

$$Ar - CH(OH)O^- + OH^- \rightleftharpoons Ar - CH(O^-)_2 + H_2O \quad \text{(Dianion, extremely electron-rich)} \tag{34}$$

3. **Step 3: Intermolecular Hydride (H^-) Transfer [RDS]:** The electron-rich dianion expels a hydride ion (H^-) directly to the carbonyl carbon of a second aldehyde molecule. This direct hydride transfer is the slow, rate-determining step:

$$Ar - CH(O^-)_2 + Ar - CHO \xrightarrow{\text{slow, RDS}} Ar - COO^- + Ar - CH_2O^- \tag{35}$$

4. **Step 4: Rapid Proton Exchange:** Alkoxide ($ArCH_2O^-$) rapidly abstracts a proton from water to yield alcohol ($ArCH_2OH$), leaving the carboxylate salt ($ArCOO^-Na^+$).

Kinetics & Crossed Cannizzaro Reaction

1. Kinetic Order:

$$\text{At moderate } [OH^-]: \quad \text{Rate} = k[\text{RCHO}]^2[OH^-] \quad \text{(Third order overall)}$$
$$\text{At high } [OH^-]: \quad \text{Rate} = k[\text{RCHO}]^2[OH^-]^2 \quad \text{(Fourth order, involving the dianion)}$$

2. Crossed Cannizzaro (Regioselectivity of Oxidation): When formaldehyde (HCHO) is mixed with another non-enolizable aldehyde (e.g., benzaldehyde, C_6H_5CHO), nucleophilic addition of OH^- occurs preferentially on formaldehyde due to its superior electrophilicity and absence of steric hindrance:

$$C_6H_5CHO + HCHO \xrightarrow{\text{conc. NaOH}} \textbf{Benzyl alcohol } (C_6H_5CH_2OH) + \textbf{Sodium formate } (HCOO^-Na^+) \quad (36)$$

Universal Rule: In crossed Cannizzaro involving formaldehyde, **formaldehyde is always oxidized to formate**, and the other aldehyde is reduced to the alcohol.

5 Carboxylic Acid Derivatives & Mechanisms

5.1 Fischer Esterification Mechanism

Acid-Catalyzed Esterification: Tetrahedral Intermediate Pathway

The reaction of a carboxylic acid with an alcohol in the presence of an inorganic acid catalyst (conc. H_2SO_4 or dry HCl) is an equilibrium addition-elimination process:

$$R - COOH + R' - OH \overset{H^+}{\rightleftharpoons} R - COOR' + H_2O \tag{37}$$

1. **Step 1: Protonation of Carbonyl Oxygen:** Protonation occurs selectively at the carbonyl oxygen (giving a resonance-stabilized cation $[R - C(OH)_2]^+$), increasing the electrophilicity of the carbonyl carbon:
$$R - COOH + H^+ \rightleftharpoons R - \overset{+}{C}(OH)_2 \tag{38}$$
2. **Step 2: Nucleophilic Attack of Alcohol [RDS]:** The alcohol oxygen lone pair attacks the activated carbonyl carbon:
$$R - \overset{+}{C}(OH)_2 + R'OH \xrightarrow{\text{slow, RDS}} R - C(OH)_2(\overset{+}{O}HR') \tag{39}$$
3. **Step 3: Proton Transfer (Tautomerism):** Proton shifts from the oxonium group to one of the hydroxyl groups:
$$R - C(OH)_2(\overset{+}{O}HR') \rightleftharpoons R - C(OH)(\overset{+}{O}H_2)(OR') \tag{40}$$
This converts a poor leaving group ($-OH$) into a good leaving group (H_2O).
4. **Step 4: Elimination of H_2O & Rearomatization/Deprotonation:** The hydroxyl oxygen lone pair assists in expelling H_2O, followed by loss of H^+ to regenerate the acid catalyst and yield the ester (RCOOR′).

Isotopic Tracer Proof of Acyl-Oxygen Bond Cleavage

When benzoic acid reacts with ^{18}O-labeled methanol ($CH_3^{18}OH$), mass spectrometry confirms that the heavy isotope ^{18}O resides entirely in methyl benzoate, and **no** ^{18}O is found in the produced water:

$$C_6H_5 - CO - OH + H - {}^{18}OCH_3 \overset{H^+}{\rightleftharpoons} C_6H_5 - CO - {}^{18}OCH_3 + H_2O \tag{41}$$

Deduction: The reaction proceeds with **Acyl-Oxygen bond fission** (R − CO—OH cleavage) in the carboxylic acid, and **Alkyl-Oxygen bond retention** (R′O—H cleavage) in the alcohol!

Steric Hindrance Hierarchy in Esterification

Because the rate-determining step involves transition from planar sp^2 (120°) to a crowded tetrahedral sp^3 intermediate (109.5°), esterification rates are governed primarily by steric bulk:

Reactivity of Carboxylic Acids: $HCOOH > CH_3COOH > RCH_2COOH > R_2CHCOOH > R_3CCOOH$

Reactivity of Alcohols: $CH_3OH >$ Primary (1°) > Secondary (2°) $\gg$ Tertiary (3°)

Note on 3° Alcohols: Tertiary alcohols (e.g., tert-butanol) do **not** react via Fischer esterification. Instead, they dehydrate rapidly to alkenes or react via alkyl-oxygen cleavage through an S_N1 carbocation intermediate (R'^+).

5.2 Decarboxylation of Carboxylic Acids

Soda-Lime Decarboxylation: Mechanism & The Carbanion Intermediate

Heating sodium or potassium salts of carboxylic acids with soda lime (a dry mixture of NaOH and CaO in a 3 : 1 mass ratio) eliminates carbon dioxide as carbonate, yielding an alkane with **one carbon atom less** than the parent carboxylate:

$$\mathrm{R-COONa + NaOH \xrightarrow[\Delta]{CaO} R-H + Na_2CO_3} \tag{42}$$

Step-by-Step Reaction Mechanism:

1. **Step 1: Dissociation & Attack by Base:** Under high thermal fusion (≈ 600 K), the carboxylate anion interacts with hydroxide ion:

$$\mathrm{R-COO^- + OH^- \rightleftharpoons [R-COO\cdots H\cdots O]^{2-}} \tag{43}$$

2. **Step 2: Heterolytic Cleavage to Free Carbanion [RDS]:** The $\mathrm{C-C}$ bond connecting the alkyl group to the carboxylate carbon undergoes heterolytic cleavage, expelling carbon dioxide (which instantly neutralizes with base to form carbonate) and generating a reactive **carbanion intermediate** ($\mathrm{R^-}$):

$$\mathrm{R-COO^- \xrightarrow{slow,\ RDS} \mathbf{R^-}\ (\textbf{Carbanion Intermediate}) + CO_2 \quad (CO_2 + 2OH^- \rightarrow CO_3^{2-} + H_2O)} \tag{44}$$

3. **Step 3: Rapid Proton Capture:** The generated carbanion instantly abstracts a proton from water or amphoteric media to form the alkane:

$$\mathrm{R^- + H_2O \xrightarrow{fast} R-H + OH^-} \tag{45}$$

Rate-Determining Rule: Because the rate-determining step generates a carbanion ($\mathrm{R^-}$), the rate of soda-lime decarboxylation is **directly proportional to the stability of the departing carbanion**:

$$\text{Decarboxylation Rate} \propto \text{Carbanion Stability: } \mathrm{CH_3^-} > 1^\circ > 2^\circ > 3^\circ \quad \text{(Reverse of carbocation stability!)} \tag{46}$$

Electron-withdrawing groups ($-I, -M$) that disperse the negative charge tremendously accelerate the rate:

$$\mathrm{O_2N-CH_2-COOH > NC-CH_2-COOH > Cl_3C-COOH > CH_3COOH} \tag{47}$$

Kolbe's Electrolytic Decarboxylation & Free Radical Coupling

When a concentrated aqueous solution of sodium or potassium carboxylate undergoes electrolysis, decarboxylative dimerization occurs at the anode:

$$\mathrm{2R-COO^-K^+ + 2H_2O \xrightarrow{Electrolysis} \mathbf{R-R} + 2CO_2\uparrow + H_2\uparrow + 2KOH} \tag{48}$$

Electrochemical Mechanism at Electrodes:

- **Anode Oxidation (Decarboxylation & Radical Dimerization):**

$$\mathrm{2R-COO^- \xrightarrow{-2e^-} 2[R-COO^\bullet]} \quad \text{(Acyloxy free radical)} \tag{49}$$

$$\mathrm{2[R-COO^\bullet] \xrightarrow{fast,\ RDS} 2R^\bullet + 2CO_2\uparrow} \quad \text{(Homolytic cleavage of C–C bond)} \tag{50}$$

$$\mathrm{2R^\bullet \xrightarrow{fast} \mathbf{R-R}} \quad \text{(Radical-radical coupling)} \tag{51}$$

- **Cathode Reduction (Evolution of $\mathrm{H_2}$ & Alkalinization):**

$$\mathrm{2H_2O + 2e^- \rightarrow \mathbf{H_2}\uparrow + 2OH^-} \tag{52}$$

Diagnostic Electrolytic Phenomena:

1. **pH Drift:** As the reaction proceeds, $\mathrm{H^+}$ is evolved as $\mathrm{H_2}$ gas while $\mathrm{OH^-}$ accumulates in solution. Hence, the **pH of the electrolyte continuously rises** (pH > 7, alkaline).
2. **Gas Ratio at Electrodes:** At the anode, $\mathrm{CO_2}$ and alkane vapor are evolved. At the cathode, pure $\mathrm{H_2}$ gas is evolved. For sodium acetate, the molar ratio of gases is (Anodic gases : Cathodic gas) = $(2\mathrm{CO_2} + 1\mathrm{C_2H_6}) : (1\mathrm{H_2}) = 3 : 1$.

Four Cardinal NEET Traps in Decarboxylation Reactions

1. **The Methane Exclusion in Kolbe Electrolysis:** Methane (CH_4) **cannot** be prepared by Kolbe electrolysis because radical coupling requires at least two alkyl radicals ($2CH_3^{\bullet} \rightarrow CH_3 - CH_3$). Potassium formate (HCOOK) electrolyzed yields H_2 and CO_2, not alkane!
2. **Radical Disproportionation By-Products:** When sodium propanoate (CH_3CH_2COONa) is electrolyzed, the intermediate ethyl radicals ($CH_3CH_2^{\bullet}$) undergo both coupling (*n*-butane) and disproportionation:
$$CH_3CH_2^{\bullet} + CH_3CH_2^{\bullet} \rightarrow CH_2 = CH_2 \text{ (Ethene)} + CH_3 - CH_3 \text{ (Ethane)} \tag{53}$$
3. **Dicarboxylic Acid Electrolysis Shortcuts:**
   - Potassium succinate ($KOOC - CH_2 - CH_2 - COOK$) yields **Ethene** ($CH_2 = CH_2$).
   - Potassium maleate or fumarate ($KOOC - CH = CH - COOK$) yields **Ethyne** ($CH \equiv CH$).
4. **Role of CaO in Soda Lime:** CaO is not a reactant. It is used because:
   - It is intensely hygroscopic and keeps solid NaOH completely dry (deliquescence prevention).
   - It prevents melting/fusion of NaOH at high temperatures, avoiding corrosion and cracking of the glass tube.

Thermal Decarboxylation of β-Keto & Gem-Dicarboxylic Acids

Carboxylic acids possessing a carbonyl group at the β-carbon (β-keto acids like acetoacetic acid, and gem-dicarboxylic acids like malonic acid) undergo spontaneous decarboxylation upon gentle heating ($\approx 373 - 423$ K) **without any added base or catalyst**:

$$R - CO - CH_2 - COOH \xrightarrow{\Delta} R - CO - CH_3 + CO_2 \uparrow \tag{54}$$

Pericyclic Mechanism via 6-Membered Cyclic Transition State: The acidic carboxylic proton forms an intramolecular hydrogen bond with the β-carbonyl oxygen. Heating triggers a concerted 6-electron cyclic electron redistribution:

$$R - CO - CH_2 - COOH \xrightarrow[\text{cyclic 6-membered T.S.}]{\Delta} R - C(OH) = CH_2 \text{ (Enol)} + CO_2 \uparrow \tag{55}$$

$$\xrightarrow{\text{tautomerism}} R - CO - CH_3 \text{ (Stable Methyl Ketone)} \tag{56}$$

The enol intermediate undergoes instantaneous, virtually irreversible keto-enol tautomerism to yield the thermodynamically favored **methyl ketone**.

5.3 Hell-Volhard-Zelinsky (HVZ) Reaction & α-Halogenation

HVZ Mechanism: In Situ Enolization via Acyl Halide

Carboxylic acids undergo selective α-halogenation when treated with bromine or chlorine in the presence of a catalytic amount of red phosphorus:

$$R - CH_2 - COOH \xrightarrow[\text{(ii) } H_2O]{\text{(i) } X_2/\text{red P}} R - CH(X) - COOH \quad (X = Cl, Br) \tag{57}$$

Step-by-Step Mechanistic Progression:

1. **Formation of Phosphorus Trihalide:** $2P + 3X_2 \rightarrow 2PX_3$.
2. **Acyl Halide Generation:** Carboxylic acids themselves enolize too slowly for direct α-halogenation. The PX_3 converts a small catalytic fraction of carboxylic acid to the corresponding acyl halide:
$$3R - CH_2 - COOH + PX_3 \rightarrow 3R - CH_2 - COX + H_3PO_3 \tag{58}$$
3. **Rapid Enolization:** The acyl halide enolizes millions of times faster than the carboxylic acid because the electronegative halogen increases the acidity of the α-protons:
$$R - CH_2 - COX \rightleftharpoons R - CH = C(OH)X \quad \text{(Enol form)} \tag{59}$$
4. **Electrophilic Halogenation:** The enol attacks X_2 to produce the α-haloacyl halide:
$$R - CH = C(OH)X + X_2 \rightarrow R - CH(X) - COX + HX \tag{60}$$

5. **Transacylation Exchange:** The α-haloacyl halide reacts with unreacted parent carboxylic acid to yield the α-halocarboxylic acid and regenerate the acyl halide, sustaining the catalytic cycle:

$$\mathrm{R-CH(X)-COX + R-CH_2-COOH \rightleftharpoons R-CH(X)-COOH + R-CH_2-COX} \tag{61}$$

HVZ Substrate Scope & Synthetic Traps

- **Requirement for α-Hydrogen:** Carboxylic acids devoid of α-hydrogens (e.g., formic acid HCOOH, benzoic acid C_6H_5COOH, and trimethylacetic acid $(CH_3)_3CCOOH$) **never** undergo the HVZ reaction.
- **Halogen Limitations:** The reaction works efficiently with Br_2 and Cl_2. Direct fluorination causes extensive fragmentation; direct iodination fails because the reverse reaction with HI dominates.
- **Downstream Synthetic Utility:** The α-halogen is readily substituted:

$$\mathrm{R-CH(Br)-COOH \xrightarrow{aq.\ KOH} R-CH(OH)-COOH} \quad (\alpha\text{-hydroxy acid})$$

$$\mathrm{R-CH(Br)-COOH \xrightarrow{excess\ NH_3} R-CH(NH_2)-COOH} \quad (\alpha\text{-amino acid})$$

6 Aromatic & Nitrogen Mechanisms

6.1 Diazotization Mechanism of Primary Aromatic Amines

Diazotization Mechanism & Nitrosonium Ion Electrophile

Primary aromatic amines react with nitrous acid at ice-cold temperature ($273 - 278$ K) to form stable arenediazonium salts:

$$\mathrm{Ar-NH_2 + NaNO_2 + 2HCl \xrightarrow{273-278\ K} Ar-N_2^+Cl^- + NaCl + 2H_2O} \tag{62}$$

Step-by-Step Reaction Sequence:

1. **Generation of Nitrosonium Ion (NO^+):** Nitrous acid is unstable and prepared in situ:

$$\mathrm{NaNO_2 + HCl \rightarrow HNO_2 + NaCl}$$

$$\mathrm{HNO_2 + H^+ \rightleftharpoons H_2\overset{+}{O} - NO \rightleftharpoons NO^+ + H_2O} \quad \text{(Nitrosonium electrophile)}$$

2. **Nucleophilic Attack:** The aniline nitrogen lone pair attacks NO^+ to form *N*-nitrosaniline cation:

$$\mathrm{Ar-NH_2 + NO^+ \rightleftharpoons Ar-\overset{+}{N}H_2-NO \xrightarrow{-H^+} Ar-NH-NO} \quad (N\text{-nitrosamine}) \tag{63}$$

3. **Proton Tautomerism & Dehydration:**

$$\mathrm{Ar-NH-N=O \rightleftharpoons Ar-N=N-OH \xrightarrow{H^+} Ar-N=N-\overset{+}{O}H_2 \xrightarrow{-H_2O} Ar-\overset{+}{N}\equiv N \longleftrightarrow Ar-\ddot{N}=\overset{+}{N}} \tag{64}$$

The diazonium ion is stabilized by resonance with the aromatic π-system.

Thermal Decomposition & Aliphatic vs Aromatic Diazonium Salts

- **Temperature Sensitivity (> 278 K/5°C):** If the temperature rises above 5°C, benzenediazonium chloride decomposes rapidly in aqueous solution:

$$\mathrm{Ar-N_2^+Cl^- + H_2O \xrightarrow{>278\ K} Ar-OH + N_2\uparrow + HCl} \quad \text{(Forms phenol quantitatively)} \tag{65}$$

- **Aliphatic Primary Amines:** Aliphatic diazonium salts ($\mathrm{R-N_2^+Cl^-}$) are exceptionally unstable even at 0°C because the alkyl group cannot disperse the positive charge by resonance. They instantaneously lose molecular N_2 to generate high-energy carbocations (R^+), yielding complex mixtures of alcohols, alkenes, and alkyl chlorides. The quantitative evolution of N_2 is utilized analytically for amino acid estimation (Van Slyke method).

6.2 Azo Dye Coupling Mechanism (Phenol vs Aniline)

Azo Coupling: Mild S_EAr & pH-Directed Chemoselectivity

Arenediazonium cations are weak electrophiles ($\mathrm{Ar-N_2^+}$) that undergo electrophilic aromatic substitution only on highly activated aromatic rings:

1. **Coupling with Phenol (Orange Dye):**
   - **Optimum pH:** Alkaline medium (**pH ≈ 9 – 10**).
   - **Mechanistic Rationale:** Phenol is a weak activator, but at pH ≈ 9 – 10 it is deprotonated into the **phenoxide ion** ($\mathrm{Ph-O^-}$), which is a vastly superior $+M$ donor. The weak diazonium electrophile attacks predominantly at the unhindered **para-position**:

$$\mathrm{Ph-N_2^+Cl^- + Ph-OH \xrightarrow[pH\ 9-10]{OH^-} \textit{p}\text{-}HO-C_6H_4-N=N-Ph\ (\textbf{Orange Dye}) + Cl^- + H_2O} \tag{66}$$

   - **High pH Hazard:** If pH > 12, the diazonium ion reacts with hydroxide to form unreactive diazotate ion ($\mathrm{Ar-N=N-O^-}$), terminating the coupling reaction.

2. **Coupling with Aniline (Yellow Dye):**
   - **Optimum pH:** Mildly acidic medium ($\mathbf{pH \approx 4-5}$).
   - **Mechanistic Rationale:** Aniline is already strongly activated. If pH < 3, aniline is protonated into the non-nucleophilic anilinium ion ($-NH_3^+$). At pH $\approx 4-5$, sufficient free aniline exists to couple at the para-position:

$$Ph - N_2^+Cl^- + Ph - NH_2 \xrightarrow[\text{pH 4–5}]{H^+} p\text{-}H_2N - C_6H_4 - N = N - Ph \ (\textbf{Yellow Dye}) + HCl \tag{67}$$

   - If the medium is basic, diazonium attacks the nitrogen lone pair instead of the ring, producing diazoaminobenzene ($Ph - N = N - NH - Ph$).

6.3 Reimer-Tiemann Reaction Mechanism

Dichlorocarbene Intermediate & Formylation of Phenol

The conversion of phenol to salicylaldehyde using chloroform and aqueous sodium hydroxide:

1. **Step 1: Generation of Dichlorocarbene ($:CCl_2$) [α-Elimination]:** Hydroxide ion deprotonates the relatively acidic proton of chloroform ($CHCl_3, pK_a \approx 15.5$ due to $-I$ of three chlorines):

$$CHCl_3 + OH^- \rightleftharpoons H_2O + {}^-:CCl_3 \quad \text{(Trichloromethyl carbanion)}$$

$${}^-:CCl_3 \xrightarrow{\text{slow, } \alpha\text{-elimination}} :CCl_2 \ \text{(Singlet Dichlorocarbene)} + Cl^-$$

   Nature of $:CCl_2$: Neutral reactive intermediate with 6 valence electrons and an empty unhybridized p-orbital $\rightarrow$ **powerful electrophile**.

2. **Step 2: Electrophilic Attack on Phenoxide:** Phenoxide attacks $:CCl_2$ predominantly at the ortho position, forming a non-aromatic cyclohexadienone carbanion intermediate:

$$C_6H_5O^- + :CCl_2 \rightarrow [o\text{-}C_6H_4(=O) - CCl_2^-] \tag{68}$$

3. **Step 3: Intramolecular Proton Shift & Rearomatization:** The aromatic proton from the ring shifts to the $-CCl_2^-$ carbon, restoring aromatic sextet stability and forming the dichloromethylphenoxide intermediate:

$$[o\text{-}C_6H_4(=O) - CCl_2^-] \rightarrow o\text{-}O^- - C_6H_4 - CHCl_2 \tag{69}$$

4. **Step 4: Alkaline Hydrolysis & Dehydration:** Hydroxide displaces both chlorine atoms via nucleophilic substitution, producing an unstable gem-diol intermediate that spontaneously loses water upon acidification to afford **salicylaldehyde** (o-hydroxybenzaldehyde).

Reimer-Tiemann Variant Trap: Chloroform vs Carbon Tetrachloride

- If **$CHCl_3$** is employed: The intermediate is a gem-dichloride ($-CHCl_2$), which hydrolyzes to give **Salicylaldehyde** (o-hydroxybenzaldehyde).
- If **CCl_4** is employed: Phenoxide attacks CCl_4 to generate a trichloromethyl intermediate ($-CCl_3$). Alkaline hydrolysis of $-CCl_3$ introduces three hydroxyl groups, which eliminate water to yield **Salicylic Acid** (o-hydroxybenzoic acid)!

6.4 Kolbe's (Kolbe-Schmitt) Reaction Mechanism

Carboxylation of Phenoxide via Carbon Dioxide Electrophile

Treatment of sodium phenoxide with carbon dioxide gas at 400 K and $4-7$ atm, followed by acidification,

produces salicylic acid:

$$C_6H_5OH \xrightarrow{NaOH} C_6H_5O^-Na^+$$

$$C_6H_5O^-Na^+ \xrightarrow[4\text{–}7 \text{ atm, } 400 \text{ K}]{CO_2} \text{Sodium Salicylate} \xrightarrow{H^+} \textbf{Salicylic Acid } (o\text{-hydroxybenzoic acid})$$

Mechanism & Ortho-Directing Sodium Chelate Effect:

1. **Electrophile:** Carbon dioxide ($O = C = O$) is a relatively weak, neutral electrophile. Un-ionized phenol is too unreactive to react with CO_2; conversion to sodium phenoxide ($Ph - O^-$) provides the requisite nucleophilic activation.
2. **6-Membered Cyclic Transition State:** The Na^+ counter-ion coordinates simultaneously to the phenoxide oxygen and one of the carbonyl oxygens of CO_2. This cyclic coordination holds CO_2 directly above the ortho-carbon, directing electrophilic attack selectively to the **ortho position**.
3. **Rearomatization:** Proton transfer from the ring carbon to oxygen regenerates the aromatic ring, followed by acidification to yield salicylic acid.

Cation Size Effect: Sodium vs Potassium Phenoxide

- **Sodium Phenoxide (Na^+):** Forms a tight 6-membered cyclic chelate due to the small ionic radius of Na^+ (102 pm), giving **salicylic acid (ortho)** as the major product ($> 80\%$).
- **Potassium Phenoxide (K^+):** K^+ has a larger ionic radius (138 pm) and cannot form an effective cyclic chelate. At 473 K, attack shifts predominantly to the less hindered para position, yielding **p-hydroxybenzoic acid** as the major product.
- **Downstream Aspirin Synthesis:**

$$\text{Salicylic acid} + (CH_3CO)_2O \xrightarrow{H^+} \textbf{Acetylsalicylic acid (Aspirin)} + CH_3COOH \tag{70}$$

6.5 Hofmann Bromamide Degradation Mechanism

Hofmann Bromamide Reaction: Concerted Intramolecular Rearrangement

Primary acid amides react with bromine in aqueous or ethanolic sodium hydroxide to yield primary amines containing **one carbon less** than the parent amide:

$$R - CONH_2 + Br_2 + 4NaOH \rightarrow R - NH_2 + Na_2CO_3 + 2NaBr + 2H_2O \tag{71}$$

Detailed Step-by-Step Mechanism:

1. **Step 1: Deprotonation & Formation of *N*-Bromoamide:** Hydroxide abstracts a proton from the amide nitrogen ($pK_a \approx 15 - 16$), generating an amide anion that attacks Br_2:

$$R - CONH_2 + OH^- \rightleftharpoons R - CON^-H + H_2O \xrightarrow{Br_2} R - CONHBr \ (N\text{-bromoamide}) + Br^- \tag{72}$$

2. **Step 2: Second Deprotonation to Bromoamide Anion:** The remaining proton on nitrogen is more acidic due to the $-I$ effect of bromine:

$$R - CONHBr + OH^- \rightleftharpoons [R - CO - \bar{N} - Br] + H_2O \tag{73}$$

3. **Step 3: Concerted 1,2-Alkyl Shift [RDS] $\rightarrow$ Formation of Isocyanate:** Expulsion of the bromide leaving group is concerted with the simultaneous intramolecular migration of the alkyl group R (with its electron pair) from the carbonyl carbon to the electron-deficient nitrogen atom:

$$[R - CO - \bar{N} - Br] \xrightarrow[\text{slow, RDS}]{-Br^-} \mathbf{R - N = C = O} \quad \textbf{(Alkyl Isocyanate Intermediate)} \tag{74}$$

   Stereochemical Consequence: The alkyl group migrates with **100% retention of stereochemical configuration** at the migrating chiral center!

4. **Step 4: Hydrolysis of Isocyanate to Carbamic Acid:** Hydroxide attacks the electrophilic carbonyl carbon of the isocyanate, followed by proton transfer to form an unstable carbamic acid:

$$R - N = C = O + OH^- \rightarrow R - NH - COO^- \tag{75}$$

5. **Step 5: Spontaneous Decarboxylation:** The carbamate undergoes spontaneous decarboxylation, releasing CO_2 (which is trapped by alkali as Na_2CO_3) and yielding the **primary amine** ($R - NH_2$).

Hofmann Bromamide Pitfalls, Stoichiometry & Reagent Trapping

- **Exact Stoichiometry:** For every 1 mol of amide, exactly **1** mol of $\mathbf{Br_2}$ and **4** mol of **NaOH** are consumed. A frequent NEET trap involves asking for the molar ratio: Amide : Br_2 : NaOH = 1 : 1 : 4.
- **Secondary and Tertiary Amides Fail:** Secondary amides ($RCONHR'$) have only one hydrogen on nitrogen and cannot form the necessary *N*-bromoamide anion; tertiary amides have zero hydrogens. Thus, only 1° amides undergo this reaction.
- **Isocyanate Trapping Proof:** If the reaction is conducted in an anhydrous alcoholic medium ($R'OH$) instead of water, the isocyanate intermediate is intercepted by alcohol to form a stable **urethane / carbamate** ($R - NH - COOR'$), proving the existence of the isocyanate intermediate!
- **Phthalimide Reaction:** Phthalimide treated with Br_2/NaOH undergoes degradation to yield **anthranilic acid** (*o*-aminobenzoic acid).

7 Master Diagnostic Matrix: Reaction Mechanisms Overview

Reaction Name	Key Intermediate	Kinetic Order	Regio / Stereochemistry	Critical NEET Exam Pitfall
Nitration of Benzene	Arenium Ion (σ-complex)	2nd order $(1+1)$	$k_H/k_D \approx 1$ (no isotope effect)	HNO_3 acts as a base; aniline gives 47% meta without $-NH_2$ protection.
Friedel-Crafts Alkylation	$1^\circ, 2^\circ, 3^\circ$ Carbocation	2nd order	Thermodynamic rearrangement	Polyalkylation occurs; deactivated rings and aniline fail.
Friedel-Crafts Acylation	Acylium Ion $(R-C\equiv\overset{+}{O})$	2nd order	No rearrangement	Deactivates ring against polyacylation; requires >1 eq. $AlCl_3$.
Halogenation of Halobenzenes	Arenium Ion ($+M$ donor)	2nd order	Ortho/Para orientation	Halogens are deactivating overall ($-I > +M$) but direct ortho/para.
S_N1 **Substitution**	Planar Carbocation (sp^2)	1st order: k[RX]	Net partial racemization + excess inversion	Rearrangements occur; hindered substrate (3°) and polar protic solvent required.
S_N2 **Substitution**	Trigonal Bipyramid T.S.	2nd order: k[RX][Nu]	100% Walden Inversion	Neopentyl halides are extremely unreactive; polar aprotic solvents accelerate.
$E1$ **Elimination**	Planar Carbocation (sp^2)	1st order: k[RX]	Saytzeff major (more substituted)	High temperature favors $E1$ over S_N1 due to $+T\Delta S^\circ$.
$E2$ **Elimination**	Anti-periplanar T.S.	2nd order: k[RX][Base]	Anti-elimination; Saytzeff vs Hofmann	Hofmann product forms with poor leaving group (F^-) or bulky base (t-BuOK).
Dehydration of Alcohols	Alkyloxonium $\rightarrow$ Carbocation	1st order in alcohol	Saytzeff alkene	Ethanol at 443 K $\rightarrow$ ethene; at 413 K $\rightarrow$ diethyl ether (S_N2).
Aldol Condensation	Enolate Carbanion $\rightarrow$ Aldol	2nd order overall	α,β-unsaturated enal/enone	Must possess α-H; cross-aldol yields 4 products unless one partner lacks α-H.
Cannizzaro Reaction	Dianion hydrate $\rightarrow H^-$ shift	3rd / 4th order	Intermolecular hydride transfer	Requires NO α-H; in crossed Cannizzaro with HCHO, HCHO is always oxidized.
Fischer Esterification	Tetrahedral Intermediate	Equilibrium	Acyl-oxygen (R – CO—OH) cleavage	^{18}O tracer remains in ester; rate hindered by bulkiness on acid/alcohol.
Soda-Lime Decarboxylation	Free Carbanion (R^-)	1st order in carboxylate	Stereochemistry cage dependent	Rate $\propto$ carbanion stability ($CH_3^- > 1^\circ > 2^\circ > 3^\circ$); CaO prevents tube fusion and deliquescence.
Kolbe Electrolysis	Alkyl Free Radical ($R^\bullet$)	2nd order radical coupling	Symmetrical R – R dimer	Methane cannot be synthesized; solution becomes strongly alkaline (pH >7) due to cathodic OH^-.
HVZ Halogenation	Enol of Acyl Halide	Catalytic in P	α-halogenation only	Must possess α-H; red P generates PX_3 which accelerates enolization.
Azo Dye Coupling	Diazonium Cation ($Ar-N_2^+$)	2nd order	Para-coupling exclusively	Phenol requires pH $\approx 9-10$ (orange); Aniline requires pH $\approx 4-5$ (yellow).
Reimer-Tiemann	Dichlorocarbene ($:CCl_2$)	2nd order	Ortho-salicylaldehyde	Intermediate is $:CCl_2$ (6 valence e^-); using CCl_4 gives salicylic acid.
Kolbe Reaction	Sodium Phenoxide chelate	2nd order	Ortho-salicylic acid (Na^+)	Na^+ gives ortho (salicylic acid); K^+ at 473 K gives para-hydroxybenzoic acid.
Hofmann Degradation	Alkyl Isocyanate ($R-NCO$)	2nd order	Retention of chiral center	Carbon chain decreases by 1; consumes 1 mol Br_2 : 4 mol NaOH; 1° amides only.